A Sparse Oblique-Manifold Nonnegative Matrix Factorization for Hyperspectral Unmixing

Ziyang Guo, Anyou Min, Bing Yang, Junhong Chen, Hong Li, *Member, IEEE*, and Junbin Gao

***Abstract*— Hyperspectral unmixing (HU) has been one of the most significant tasks in hyperspectral image (HSI) processing. In recent years, nonnegative matrix factorization (NMF) has received great attention in the HU due to its simultaneous estimation, flexible modeling, and little requirement on prior information. However, several common NMF algorithms still suffer from high computational complexity, instability, and low convergence rate. Motivated by the matrix manifold theory, this article proposes a new sparse oblique-manifold ($\mathcal{OB}$) NMF method from the perspective of matrix manifold. The critical idea of the proposed method is to regard the abundance matrix as locating on the oblique manifold, which eliminates its constraint of nonnegativity and sum-to-one and incorporates its intrinsic Riemannian geometry. Meanwhile, the $L_{1/2}$-norm on the Euclidean space can be transformed equivalently into the L_1-norm on oblique manifold. Then, via solving this sparse $\mathcal{OB}$NMF by the Riemannian conjugated gradient (RCG) algorithm and the multiplicative iterative rule, the proposed method not only ensures improvement in the solution accuracy but also leads to a much faster convergence rate. Experimental results from the synthetic and real-world datasets illustrate the effectiveness and efficiency of the proposed method compared with the state-of-the-art NMF methods in HU.**

***Index Terms*— $L_{1/2}$ regularization, hyperspectral unmixing (HU), nonnegative matrix factorization (NMF), oblique manifold.**

I. Introduction

HYPERSPECTRAL images (HSIs) acquired by hyperspectral imaging sensor are often a mixing of the spectra from different substances. A fundamental task for hyperspectral image analysis is hyperspectral unmixing (HU), which aims to decompose all pixels into an endmember matrix and their corresponding abundances. Due to the high spectral resolution and low spatial resolution of HSI, HU has been one of the most significant tasks in recent HSI research [1]–[6]. In general, existing HU technologies can be classified into two groups by the model type: linear mixing model (LMM)-based and nonlinear mixing model (NLMM)-based. The NLMM [7] often considers the mutual influence of different or the same signal materials or the intimate mixing in microscale, which makes the assumption varies a lot and the problem more complicated to deal with. Thus, in most studies, LMM is utilized due to its mathematical simplicity, interpretability, and satisfactory performance. Recent HU methods on LMM can be generally divided into five categories: sparsity regression-based, geometry-based, intelligent optimization-based, deep learning-based, and statistics-based.

Sparse-regression-based methods assume that pixels consist of the pure spectra existed in the known spectral libraries in advance. Thus, it aims to find an optimal linear combination of endmembers. Fu *et al.* [8] accomplished a sparse regression HU based on self-dictionary multiple measurement vector (SD-MMV), which treats the observed pixels as the dictionary. However, this method relies much on an exact prior dictionary of endmembers. Geometry-based research usually has a "pure pixel" assumption, which ensures the existence of the pure pixel, and it includes representative methods, such as vertex component analysis (VCA) [9], pixel purity index (PPI) [10], N-Findr [11], and minimum volume-based methods. Minimum volume-based methods, such as simplex identification via split augmented Lagrangian (SISAL) [12] and minimum volume simplex analysis (MVSA) [13], are proposed to seek for endmember matrix from the perspective of the simplex. However, this assumption is often not suitable in real scenarios. Spectral variability has shown to be significant in estimating the abundance map, which is derived from illumination conditions or atmosphere or intrinsic variability [14], [15]. Hong *et al.* [16] applied a data-driven learning strategy in inverse problems of HU to deal with the spectral variability. Hong and Zhu [17] introduced a subspace unmixing with low-rank attribute embedding to deal with spectral variability in inverse problems of HU.

Recently, a series of intelligent optimization-based endmember extraction methods has been proposed for endmember extraction in HU research, due to its superiority to solve the combinatorial optimization problem and the multiobjective optimization problem. Liu *et al.* [18] introduced a quantum-behaved particle swarm optimization (QPSO) algorithm to improve the performance of endmember extraction from the perspective of the quantum theory. Du *et al.* [19] gave an updated version of QPSO algorithm, which can theoretically guarantee the convergence of the algorithm. Liu and Zhu [20] developed multiobjective particle swarm optimization for the identification of multiple endmember spectra with variability.

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Deep learning methods have gradually received attention nowadays, due to its excellent performance. Savas *et al.* [21] proposed a two-staged autoencoder network by introducing additional layers and an inner product to achieve an optimal solution. Deng *et al.* [22] proposed a deep belief network to study the joint probabilities in land use and land cover, which ensures better object-based identification for HU.

Nowadays, the statistics-based method can be a powerful method for HU problem from the statistical perspective. There exist several popular algorithms, including independent component analysis (ICA) [23], nonnegative matrix factorization (NMF), and metropolis-within-Gibbs sampler [24], [25]. Zare *et al.* [24] represented a metropolis-within-Gibbs sampler for piecewise convex HU and endmember extraction. Zhou *et al.* [25] adopted Gaussian mixture models to represent endmember variability since the distribution of a material is often not Gaussian in real scenarios. NMF has been widely used due to its effectiveness, flexible modeling, and less requirement on prior information. NMF is initially proposed by Lee and Seung [26], who intended to decompose a nonnegative matrix into two nonnegative matrices simultaneously. Thus, under this method for HU, the two achieved matrices can be represented as the abundance matrix and the spectral endmember matrix. Note that no subtraction is applied in the multiplicative iterative rule of NMF.

There exist several NMF-based methods for HU. L_1-NMF [27] puts the sparsity constraint into the objective function to add the sparsity of abundance matrix and enhance the performance of HU. $L_{1/2}$-NMF [28] learns the best L_q regularizer that can be used in the HU, but at the expense of larger computational complexity than L_1-NMF. Lu *et al.* [29] incorporated manifolding regularization into sparsity-constrained NMF to keep the geometric similarity between the original image and the abundance maps. Shu *et al.* [30] proposed a multilayer manifold sparse NMF on both spectral signature matrix and the abundance matrix. Li *et al.* [31] incorporated the $L_{2,1}$-norm into NMF, not only to estimate the optimal endmember number but also to get good row sparsity. Now nonnegative tensor factorization (NTF) [32] has been developed by using a tensor theory to incorporate spatial information of HSI. Several existing NMF algorithms tend to use manifold learning as regularization to explore the intrinsic structure of the data. Peng *et al.* [33], [34] proposed a self-paced regularizer to improve the robustness of the NMF model to different levels of noises. Yao *et al.* [35] built a nonlocal HSI total variation regularizer and a log-sum regularizer in the NMF model to characterize the diverse spatial distribution of different endmembers. As is shown, a lot of NMF methods tend to add different kinds of regularization to ensure better properties for abundance matrix or endmember matrix, such as the sparsity, low rank, and local information, and few attention has been paid on exploring the latent structure of the endmember matrix or the abundance matrix.

Inspired by the matrix manifold optimization theory [36]–[38], we intend to explore the embedded submanifold of the HSI data space. Absil and Gallivan [39] proposed a joint diagonalization-based method on $\mathcal{OB}$ for ICA by a Riemannian trust-region approach. Selvan *et al.* [40] showed the superiority of $\mathcal{OB}$ algorithms for source-adaptive ICA contrast problem than other Riemannian and Euclidean approaches. Kleinsteuber and Shen [41] built a minimization problem for blind source seperation on $\mathcal{OB}$. Oblique manifold is an important manifold, which is a product manifold of a series of spheres. Classical constrained optimization techniques work in an embedded space with a much larger dimension than in the manifold. Optimization algorithms on the manifold often have a lower complexity and quite often have better numerical properties [39]. Most studies for oblique manifold are based on ICA. Due to the connection between ICA and NMF in real applications, we explore to extend the research of oblique manifold on NMF problems.

In this article, we incorporate the superiority of $L_{1/2}$-NMF and extend it to a new perspective by regarding the abundance matrix as lying on a specific lower dimensional manifold, i.e., oblique manifold. As we know, LMM for HU often considers two structural constraints, abundance nonnegative constraint (ANC) and abundance sum-to-one constraint (ASC), which ensures physical interpretation of the model. By regarding the abundance matrix as lying on the oblique manifold, the ANC and ASC constraints of the abundance matrix can be eliminated. Meanwhile, the $L_{1/2}$-norm on the Euclidean space can be transformed into the L_1-norm on the oblique manifold without damaging the effectiveness of $L_{1/2}$-norm. Consequently, lower computational complexity can be derived. Specifically, since each entry in the abundance matrix will be the square of a new number (negative or positive), all entries will always be nonnegative, and then, no extra nonnegativity is needed. Daniel and Chang [42] proposed a fully constrained least square (FCLS) method to adopt the ASC, which is adopted by most studies who take ASC constraint into account. Instead of utilizing FCLS, we integrate the ASC constraint into the condition of oblique manifold. In this manner, no constraint has remained in the loss function, and this can be regarded as a novel way to deal with ASC and ANC. Moreover, we proposed a Riemannian conjugated gradient (RCG) method on the oblique manifold, with accelerated convergence rate. The corresponding algorithm analysis and experiments will be given in Sections III and IV.

The rest of this article is organized as follows. Section II shows the basic knowledge of the LMM, $L_{1/2}$-NMF model, and the geometry of oblique manifold. Section III presents the proposed L_1-$\mathcal{OB}$NMF model, its updating algorithm, and the algorithm analysis. Section IV shows the experiments on synthetic data and real-world hyperspectral dataset. The conclusion is given in Section V.

II. Preliminary

A. Linear Mixed Model

In the unmixing scheme, the LMM is adopted to derive the spectrum of a pixel and their respective abundances, with M bands and K endmembers in the observed data. For a given pixel vector $\boldsymbol{x} = [x_1, \ldots, x_M]^T$, M denotes the number of the spectral bands. The original model can be given as $\boldsymbol{x} = \mathbf{A}\mathbf{s} + \varepsilon$, in which $\mathbf{A} = [\mathbf{a}_1, \ldots, \mathbf{a}_L] \in \mathbb{R}^{M \times L}$ is the endmember matrix,

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$\mathbf{s} = [s_1, \ldots, s_L]^T \in \mathbb{R}^L$ is the abundance vector of a pixel, and ε is an $M \times 1$ vector of the noise. In LMM, $\mathbf{s}$ satisfies the ANC (i.e., $s_l \geq 0$) and the ASC (i.e., $\sum_{i=1}^{L} s_l = 1$) constraints. By a matrix form, for N pixels, the LMM can rewritten as

$$\mathbf{X} = \mathbf{AS} + \mathbf{E} \tag{1}$$

where $\mathbf{X} \in \mathbb{R}^{M\times N}$, $\mathbf{S} \in \mathbb{R}^{L\times N}$, and $\mathbf{E} \in \mathbb{R}^{M\times N}$ represent the observed data, the endmember abundances, and the noise, respectively. In most cases, the noise $\mathbf{E}$ is usually assumed to be independent and identically distributed Gaussian noise.

B. NMF With $L_{1/2}$ Constraint

To obtain a sparser solution of $\mathbf{A}$ and $\mathbf{S}$, the loss function for NMF with an $L_{1/2}$ regularizer was first explored by Qian *et al.* [28] to ensure sparsity and computational efficiency. Then, the $L_{1/2}$-NMF model is defined by

$$f(\mathbf{A}, \mathbf{S}) = \frac{1}{2}\|\mathbf{X} - \mathbf{AS}\|_2^2 + \lambda\|\mathbf{S}\|_{1/2} \tag{2}$$

where $\|\mathbf{S}\|_{1/2} = \sum_{l,n=1}^{L,N} s_n(l)^{1/2}$ and $s_n(l)$ represents the abundance component of lth endmember for the nth pixel in the HSI. One widely used optimization algorithm for this model is multiplicative iterative algorithm which applies a multiplicative update (MU) rule in NMF as follows:

$$\mathbf{A} \leftarrow \mathbf{A} \odot \mathbf{XS}^T./\mathbf{ASS^T} \tag{3}$$

$$\mathbf{S} \leftarrow \mathbf{S} \odot \mathbf{A}^T\mathbf{X}./\left(\mathbf{ASS^T} + \frac{\lambda}{2}\mathbf{S}^{-1/2}\right). \tag{4}$$

C. Oblique-Manifold Geometry

We restrict the abundance matrix S onto the oblique manifold

$$\mathcal{OB}(L, N) := \left\{\mathbf{B} \in \mathbb{R}^{L\times N} \middle| r(\mathbf{B}) = L, \operatorname{ddiag}\left(\mathbf{B}^T\mathbf{B}\right) = I_N\right\} \tag{5}$$

in which I_N denotes the $(N \times N)$ identity matrix, $r(\mathbf{B})$ is the rank of $\mathbf{B}$, and ddiag(Z) forms a diagonal matrix, and its entries are entries of Z in the same position. It can be seen as a relaxed version of a widely used stiefel manifold, whose definition is $\mathcal{ST}(L, N) := \{\mathbf{B} \in \mathbb{R}^{L\times N} | r(\mathbf{B}) = L, (\mathbf{B}^T\mathbf{B}) = I_N\}$. Note that $\mathcal{OB}$ is isometric to the product of N unit spheres in $\mathbb{R}^L$ with a Riemannian structure. Thus, the gradients of optimization on $\mathcal{OB}$ can be derived first for the unit sphere $\mathcal{S}^{L-1}$ and extended to $\mathcal{OB}(L, N)$. The tangent space to $\mathcal{OB}$ at $\mathbf{B}$ is

$$T_{\mathbf{B}}\mathcal{OB} = \left\{\mathbf{Z} : \operatorname{ddiag}\left(\mathbf{B}^T\mathbf{Z}\right) = 0\right\} \tag{6}$$

which means that $\mathbf{B}_i^T\mathbf{Z}_i = 0$, $i = 1, \ldots, N$, where $\mathbf{B}_i$ denotes the ith column of $\mathbf{B}$ and $\mathbf{Z}_i$ denotes the ith column of $\mathbf{Z}$. The dimension of $\mathcal{OB}$ is $\dim(\mathcal{OB}) = N(L-1)$. The normal space is

$$N_{\mathbf{B}}\mathcal{OB} = \left\{\mathbf{B}D : \ D \in \mathbb{R}^{N\times N} \text{ diagonal}\right\}. \tag{7}$$

Projections of $\mathbf{Z} \in \mathbb{R}^{L\times N}$ into the normal and tangent spaces are

$$P_{N_{\mathbf{B}}}(\mathbf{Z}) = \mathbf{B}\operatorname{ddiag}\left(\mathbf{B}^T\mathbf{Z}\right) \tag{8}$$

$$P_{T_{\mathbf{B}}}(\mathbf{Z}) = \mathbf{Z} - \mathbf{B}\operatorname{ddiag}\left(\mathbf{B}^T\mathbf{Z}\right). \tag{9}$$

A retraction is defined as

$$R_{\mathbf{B}}(\mathbf{Z}) = (\mathbf{B} + \mathbf{Z})\left(\operatorname{ddiag}\left((\mathbf{B} + \mathbf{Z})^T(\mathbf{B} + \mathbf{Z})\right)\right)^{-1/2} \tag{10}$$

which provides a correspondence between tangent vectors and elements on the manifold. It aims to add $\mathbf{Z}$ to $\mathbf{B}$ and scale the columns of the sum. The gradient of f at $\mathbf{B}$ represented by $\operatorname{grad} f(\mathbf{B})$ is defined as the element of $T_{\mathbf{B}}\mathcal{OB}$ such that $\langle \operatorname{grad} f(\mathbf{B}), \xi\rangle = \operatorname{Df}(\mathbf{B})[\xi]$ for all ξ in $T_{\mathbf{B}}\mathcal{OB}$, where $\operatorname{Df}(\mathbf{B})[\xi]$ represents the directional derivative of f in the direction of ξ.

In each iteration of the optimization, the next iteration direction determined by the tangent space $T_{\mathbf{B}_k}\mathcal{S}$ should be transported to the tangent space $T_{\mathbf{B}_{k+1}}\mathcal{S}$. Since the manifold is endowed with a retraction R, a vector transport for $\mathcal{S}^{L-1}$ is given by

$$\mathcal{T}_{\xi_k}(\lambda_k) = \left[I - \frac{(\mathbf{B}_k + \xi_k)(\mathbf{B}_k + \xi_k)^T}{\|\mathbf{B}_k + \xi_k\|_2^2}\right]\lambda_k \tag{11}$$

which can be applied on $\lambda_k \in T_{\mathbf{B}_k}\mathcal{S}$ from (11). The vector transport and retraction are also applicable on the oblique manifold.

III. Algorithm for L_1-$\mathcal{OB}$NMF Unmixing

A. Sparse $\mathcal{OB}$-Based NMF

As mentioned earlier, the objective function of $L_{1/2}$-NMF is presented as follows:

$$g(\mathbf{X}) = \arg\min_{\mathbf{A},\mathbf{S}} \frac{1}{2}\|\mathbf{X} - \mathbf{AS}\|_F^2 + \lambda\|\mathbf{S}\|_{1/2} \quad \text{s.t. } \mathbf{S} \geq 0, \quad 1_L^T\mathbf{S} = 1_N^T \tag{12}$$

where the notation $\|\cdot\|_F$ is the Frobenius norm and 1_L and 1_N denote all one column vectors of size L and N, respectively. Inspired by the knowledge of compositional data analysis [43], [44], we consider the ASC constraint from the perspective of longitudinal compositional data

$$\mathbf{s}_i \in \left\{\left[s_{1,i}, \ldots, s_{L,i}\right]^T | s_{l,i} \geq 0, l = 1, \ldots, L; \ i = 1, \ldots, N; \sum_{l=1}^{L} s_{l,i} = 1\right\}. \tag{13}$$

Then, we apply root transformations to the longtitudinal compositional data S, obtaining

$$\mathbf{b}_i = \left[b_{1,i}, \ldots, b_{L,i}\right]^T = \left[\pm s_{1,i}^{1/2}, \ldots, \pm s_{L,i}^{1/2}\right]^T \tag{14}$$

and thus, $\mathbf{B} = [\mathbf{b}_1, \ldots, \mathbf{b}_N]$, i.e., $\mathbf{B} \odot \mathbf{B} = \mathbf{S}$. There is no problem with zeros as with some other proposed transformations for $\mathbf{B}$. Note that oblique manifold is a submanifold of Cartesian product of spheres, it is then a natural approach to consider $\mathbf{B}$ on the oblique manifold. The ASC constraint will be transformed into the condition of an oblique manifold. Considering the definition of $\mathcal{OB}$, $\mathbf{B}$ can be regarded as lying on an $\mathcal{OB}$ by $\operatorname{ddiag}(\mathbf{B}^T\mathbf{B}) = I_N$. Therefore, the constraint of ASC can be dimissed. At the same time, the nonnegativity of $\mathbf{S}$ can be dismissed. Then, the loss function can be transformed as L_1-$\mathcal{OB}$NMF as follows:

$$g(\mathbf{X}) = \arg\min_{\mathbf{A},\mathbf{B}} \frac{1}{2}\|\mathbf{X} - \mathbf{A}(\mathbf{B} \odot \mathbf{B})\|_F^2 + \lambda\|\mathbf{B}\|_1 \quad \text{s.t. } \mathbf{B} \text{ lying on an } \mathcal{OB} \tag{15}$$

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where $\odot$ is the hadamard product and $|\mathbf{B}|$ is the absolute value of $\mathbf{B}$.

B. Updating Rules

Similar to the classical NMF, the cost function f is not convex with respect to $\mathbf{A}$ and $\mathbf{B}$ jointly, and thus, a multiplicative iterative algorithm to find a global minima is required. Considering the Lagrange operator $\mathcal{L}$ as

$$\begin{aligned}\mathcal{L} &= \arg\min_{\mathbf{A},\mathbf{B}} \frac{1}{2}\|\mathbf{X}-\mathbf{A}(\mathbf{B}\odot\mathbf{B})\|_F^2 + \lambda\|\mathbf{B}\|_1 \\ &= \frac{1}{2}\mathrm{tr}\big(\mathbf{X}\mathbf{X}^T\big) - \mathrm{tr}\big(\mathbf{X}(\mathbf{B}\odot\mathbf{B})^T\mathbf{A}^T\big) \\ &\quad + \frac{1}{2}\mathrm{tr}\big(\mathbf{A}(\mathbf{B}\odot\mathbf{B})(\mathbf{B}\odot\mathbf{B})^T\mathbf{A}^T\big) + \lambda\|\mathbf{B}\|_1 \end{aligned} \tag{16}$$

where the derivative of the Hadamard product $\odot$ is $d(\mathbf{B}\odot\mathbf{B}) = 2d(\mathbf{B})\odot\mathbf{B}$. Then, first taking the partial derivative of $\mathcal{L}$ with respect to $\mathbf{A}$ yields

$$\frac{\partial\mathcal{L}}{\partial\mathbf{A}} = -\mathbf{X}(\mathbf{B}\odot\mathbf{B})^T + \mathbf{A}(\mathbf{B}\odot\mathbf{B})(\mathbf{B}\odot\mathbf{B})^T + \Psi. \tag{17}$$

Due to the Karush–Kuhn–Tucker condition $\Psi_{lk}A_{lk} = 0$, then the updating rule of $\mathbf{A}$ can be required as

$$\mathbf{A} \leftarrow \mathbf{A}\odot\mathbf{X}(\mathbf{B}\odot\mathbf{B})^T./\mathbf{A}(\mathbf{B}\odot\mathbf{B})(\mathbf{B}\odot\mathbf{B})^T. \tag{18}$$

The Euclidean gradient $\nabla\mathcal{L}(\mathbf{B})$ of $\mathcal{L}$ with respect to $\mathbf{B}$ is evaluated using (4) as

$$\nabla\mathcal{L}(\mathbf{B}) = \big(-2\mathbf{A}^T\mathbf{X} + 2\mathbf{A}^T\mathbf{A}(\mathbf{B}\odot\mathbf{B})\big)\odot\mathbf{B} + 0.5\lambda\cdot\mathrm{sgn}(\mathbf{B}) \tag{19}$$

in which $\mathrm{sgn}(\cdot)$ is the sign function.

Since $\mathbf{B}$ lies on the $\mathcal{OB}$, the projection of $\nabla\mathcal{L}(\mathbf{B})$ onto the tangent space to $\mathcal{OB}$ at $\mathbf{X}$ gives the Riemannian gradient $\mathrm{grad}\mathcal{L}(\mathbf{B})$, which is computed as

$$\begin{aligned}\mathrm{grad}\mathcal{L}(\mathbf{B}) &= P_{T_\mathbf{B}}(\nabla\mathcal{L}(\mathbf{B})) \\ &= \nabla\mathcal{L}(\mathbf{B}) - \mathbf{B}\cdot\mathrm{ddiag}\big(\mathbf{B}^T\nabla\mathcal{L}(\mathbf{B})\big).\end{aligned} \tag{20}$$

With $\mathrm{grad}\mathcal{L}(\mathbf{B})$, we are able to calculate the search direction $\Xi_k \in \mathcal{T}_{\mathbf{B}_k}\mathcal{OB}$, $\Xi_{k+1} \in \mathcal{T}_{\mathbf{B}_{k+1}}\mathcal{OB}$ and the step size $a_k > 0$ in Section III-C. In the following, $\mathbf{B}_k$ denotes the kth iteration of $\mathbf{B}$.

After getting Ξ_k and a_k, the updating of $\mathbf{B}$ can be given via a retraction, which constrains $\mathbf{B}_{k+1}$ on the $\mathcal{OB}$. Since oblique manifold can also be regarded as a submanifold of the Cartesian product of spheres, it shares some properties with spheres. When optimizing on $\mathcal{S}^{L-1}$, the kth iterate moves $\mathbf{B}_k$ along the direction Ξ_k. Thus, a retraction on the $\mathcal{OB}$, which ensured the next iterate $\mathbf{B}_{k+1}$ to retain on the $\mathcal{OB}$, is given via (10) as

$$R_{(\mathbf{B}_k)_i}(a_k\Xi_k) := (\mathbf{B}_k + a_k\Xi_k)_i/\|(\mathbf{B}_k + a_k\Xi_k)_i\| \tag{21}$$

where $\|\cdot\|$ denotes the Euclidean norm and $(\mathbf{B}_k)_i$ denotes the ith column of $\mathbf{B}_k$. Then, the updating of $\mathbf{B}$ on $\mathcal{OB}$ can be given by

$$\mathbf{B}_{k+1} \leftarrow R_{\mathbf{B}_k}(a_k\Xi_k) \tag{22}$$

as shown in Fig. 1.

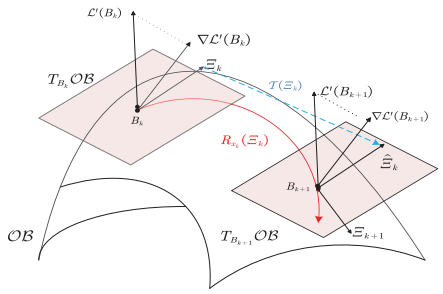


Fig. 1. Retraction and vector transport on $\mathcal{OB}$.

C. Conjugate Gradient Algorithm on $\mathcal{OB}$

Since $\mathbf{B}$ lies on the oblique manifold, the multiplicative rule in conventional NMF is no longer applicable. Here, we adopt the RCG method to accelerate the convergence rate of $\mathbf{B}$. The conjugate gradient (CG) method is popular by its low memory requirement and strong local and global convergence properties [45]. Also, it has fast convergence without computing the Hessian of the function. Thus, we adopt the Riemannian CG method for the iteration of $\mathbf{B}$. The algorithm finds a solution by finding the conjugated directions. Nonlinear CG methods and linear CG share the same idea, but nonlinear CG works for nonlinear functions and the Riemannian CG method needs one more step for parallel transport further.

Since the tangent space gradient $\mathrm{grad}\mathcal{L}(\mathbf{B})$ on $\mathcal{OB}$ at $\mathbf{B}$ is computed, the search directions Ξ_{k+1} can be derived. To ensure global convergence here, the choice of a_k requires an exact line search and a_k satisfies the weak Wolfe conditions

$$\mathcal{L}(\mathbf{B}_{k+1}) \le \mathcal{L}(\mathbf{B}_k) + c_1 a_k(\mathrm{vec}(\mathrm{grad}\mathcal{L}(\mathbf{B}_k)))^T\mathrm{vec}(\Xi_k) \tag{23}$$

$$(\mathrm{vec}(\mathrm{grad}\mathcal{L}(\mathbf{B}_{k+1})))^T\hat{\Xi}_k \le c_2(\mathrm{vec}(\mathrm{grad}\mathcal{L}(\mathbf{B}_k)))^T\mathrm{vec}(\Xi_k) \tag{24}$$

where $0 < c_1 < c_2 < 1$ (default $c_1 = 0.0001$ and $c_2 = 0.9$ [40]) and $\mathrm{vec}(\cdot)$ means to vectorize a matrix. Equation (23) is called the Armijo condition, which is often used to stipulate a_k to give sufficient decrease. Also, the initialization of a_k is given in [45] by

$$a_{kinit} = \frac{a_{k-1}(\mathrm{vec}(\mathrm{grad}\mathcal{L}(\mathbf{B}_{k-1})))^T vec(\Xi_{k-1})}{(\mathrm{vec}(\mathrm{grad}\mathcal{L}(\mathbf{B}_k)))^T vec(\Xi_k)}. \tag{25}$$

The Armijo condition can guarantee sufficiently small values of a, but not be enough to ensure the algorithm to have good efficiency. As to prevent very short step, the curvature condition is given by (24). Moreover, the search direction Ξ_{k+1} for the next iteration also needs to be determined. In Euclidean cases, we have

$$\Xi_{k+1} = -\nabla\mathcal{L}(\mathbf{B}_{k+1}) + \beta_{k+1}\Xi_k \tag{26}$$

where β_{k+1} denotes the update parameter. However, both $\mathrm{grad}\mathcal{L}(\mathbf{B}_{k+1})$ and Ξ_{k+1} correspond to the tangent space $T_{\mathbf{B}_{k+1}}\mathcal{OB}$ at $\mathbf{B}_{k+1}$, while Ξ_k belongs to $T_{\mathbf{B}_k}\mathcal{OB}$ at $\mathbf{B}_k$. Smith [46] used the parallel translation in each iteration to generalize the CG method in this manifold situation. Recently, vector transport [47], [48] has been a computational efficient

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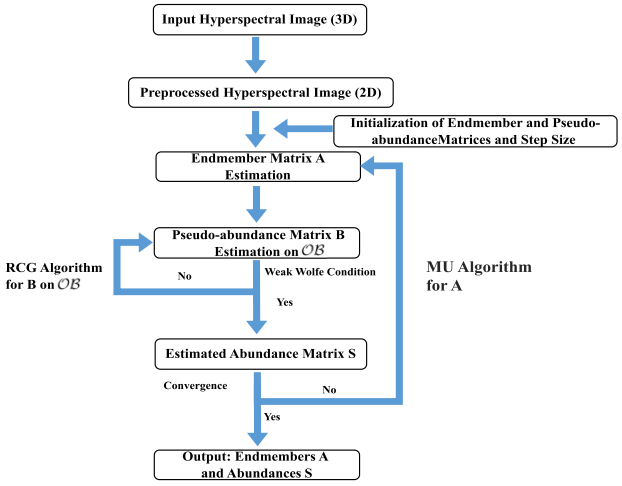


Fig. 2. Workflow of Algorithm 1.

manner to replace the parallel translation. A possible vector transport

$$\mathcal{T}_{a_k\Xi_k}(\Xi_k) = \left[I - \frac{(\mathbf{B}_k + a_k\Xi_k)(\mathbf{B}_k + a_k\Xi_k)^T}{\|\mathbf{B}_k + a_k\Xi_k\|^2}\right]\Xi_k \quad (27)$$

is defined to transport a tangent vector Ξ_k along the geodesic to the tangent space $T_{\mathbf{B}_{k+1}}\mathcal{OB}$, and then, $\hat{\Xi}_k = \text{vec}(\mathcal{T}_{a_k\Xi_k}(\Xi_k)) \in T_{\mathbf{B}_{k+1}}\mathcal{OB}$. Then, it is natural to obtain Ξ_{k+1} in the manifold case, which is illustrated as

$$\Xi_{k+1} = -\text{grad}\mathcal{L}(\mathbf{B}_{k+1}) + \beta_{k+1}\text{unvec}\big(\hat{\Xi}_k\big). \quad (28)$$

The choice of β_{k+1} will affect the convergence properties. A lot of methods have been given to the choice of β_{k+1}. A hybrid strategy of β_{k+1} proposed by Dai and Yuan [50] is expressed as

$$\beta_{k+1} = \max\big\{0, \min\big(\beta_{k+1}^{\text{HS}}, \beta_{k+1}^{\text{DY}}\big)\big\}. \quad (29)$$

The Hestenes–Stiefel (HS) formulas are given by $\beta_{k+1}^{\text{HS}} = ((\text{vec}(\text{grad}\mathcal{L}(\mathbf{B}_{k+1})))^T y_k)/(\hat{\Xi}_k^T y_k)$ in [49]; and the Dai–Yuan (DY) formulas are given by $\beta_{k+1}^{\text{DY}} = (\|\text{vec}(\text{grad}\mathcal{L}(\mathbf{B}_{k+1}))\|^2)/(\hat{\Xi}_k^T y_k)$ in [50], in which y_k is denoted as

$$y_k = \text{vec}\big(\text{grad}\mathcal{L}(\mathbf{B}_{k+1}) - \mathcal{T}_{a_k\Xi_k}(\text{grad}\mathcal{L}(\mathbf{B}_k))\big). \quad (30)$$

The HS formulas have strong convergence properties but might have a tiny step and bad search direction; The DY formulas often have a better performance than HS formulas but might suffer from poor convergence. Thus, a hybrid strategy [51] has shown to be more effective.

A workflow in Fig. 2 is given to illustrate the entire process of algorithm 1. It shows that after initialization of the endmember matrix **A**, pseudo-abundance matrix **B**, and step size in the RCG algorithm, **A** and **B** are updated by MU and RCG, individually and alternatively. During the updating of **B**, the derived a_k has to satisfy the weak Wolfe condition before reaching the next iteration. Finally, the algorithm 1 stops when it converges.

Algorithm 1 L_1-$\mathcal{OB}$NMF

Input: X, AInit, BInit, λ, tol
Output: Estimated endmember matrix **A** and abundance matrix **S**;

Initialize **A**, **B** and λ with default or random values, $k = 0$;
Evaluate $f(\mathbf{B}_0)$, grad$(f(\mathbf{B}_0))$ and Ξ_0;
Determine a_0 by $a_0 = 1$;
repeat
  Update **A** by applying $\mathbf{A} \leftarrow \mathbf{A} \odot \mathbf{S}^T./\mathbf{A}\mathbf{S}^\mathbf{T}$;
  $a_k = a_{\text{init}}$ by (25), compute $\mathbf{B}_{k+1}$ and $\mathcal{L}(\mathbf{B}_{k+1})$ by (22) and (16);
  repeat
    else choose $a_k = \gamma * a_{\text{kguess}}$;
    Obtain $\mathbf{B}_{k+1}$ and $f(\mathbf{B}_{k+1})$;
  until a_k *satisfies the weak Wolfe conditions,*;
  Evaluate grad$\mathcal{L}(\mathbf{B}_{k+1})$ and deduce y_k and β_{k+1} by (20), (29), and (30) respectively;
  Find Ξ_{k+1} via (28) for the next iteration;
  Update $\mathbf{S} = \mathbf{B}_{k+1} \odot \mathbf{B}_{k+1}$;
  $k \leftarrow k + 1$;
until $k >$ *maxIterNumber or* $\|\mathcal{L}_{k+1} - \mathcal{L}_k\|_2^2 < \varepsilon$ *or* $(\|\text{vec}(\text{grad}\mathcal{L}(\mathbf{B}_{k+1}))\|_\infty)/((1 + \|\text{vec}(\text{grad}\mathcal{L}(\mathbf{B}_0))\|_\infty)) < \epsilon$;
return A,**S**.

D. Algorithm Analysis

1) Implementation Issues: There are four issues to be discussed when implementing the proposed algorithm. The first is the initialization of **A** and **B**. As most NMF algorithms for HU do, random initialization, Nfindr-FCLS initialization, and VCA-FCLS initialization are used in this article. In the random initialization, initial **A** randomly chooses the columns from the observation and initial **B** randomly chooses any values between -1 and 1 to be its entries. VCA and Nfindr are the two common methods to initialize **A**. FCLS can be adopted to initialize **B**.

The second issue is the coefficient λ of sparsity regularization. As in [28] and [52], we give an estimation of λ by cross validation.

Third, we discuss the stopping criteria. For the iteration of **B**, the iteration terminates when satisfying

$$\frac{\|\text{vec}(\text{grad}\mathcal{L}(\mathbf{B}_{k+1}))\|_\infty}{(1 + \|\text{vec}(\text{grad}\mathcal{L}(\mathbf{B}_0))\|_\infty)} < \epsilon. \quad (31)$$

Also, the whole iteration stops when the iteration number reaches 1000 and the deviation of loss between the formal and latter iterations reaches 10^{-4}, i.e., $\|\mathcal{L}_{k+1} - \mathcal{L}_k\|_2^2 < 10^{-4}$.

Finally, since our proposed algorithm does not relate to the choice of endmember number, the number of endmembers in this article will be given by prior information from several existing algorithms, such as virtual dimensionality (VD) [53], HySime [54], or R-CoNMF [31].

2) Computational Complexity Analysis: In this section, the computational complexity is analyzed in detail. The calculation time of each iteration is given in Table I.

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TABLE I
FLOATING-POINT CALCULATION TIMES FOR EACH ITERATION IN L_1-$\mathcal{OB}$NMF

	Update A	Update S	Total
Addition	$MNL+(M+N)L^2$	$MNL+(M+N)L^2+(2+3t)NL$	$2MNL+2(M+N)L^2+(2+3t)NL$
Multiplication	$MNL+(M+N)L^2$	$((M+11t+9)NL+(M+2N+2tN)L^2$	$(2M+11t+9)NL+(2M+3N+2tN)L^2$
Division	ML	$2NL$	$ML+2NL$
Subtraction	–	$NL+(4NL+L^2)t$	$NL+(4NL+L^2)t$

It is natural to see that that the computational complexity in calculating **A** in the proposed method is the same as that in the standard NMF method [28]. Thus, we mainly turn to see that in calculating **s** and count the time of floating-point calculation in each iteration. From [29], the cost of $\mathbf{S}^{-1/2}$ is $(\mathrm{LN})^2$. In our model, the cost of $\lambda\|\mathbf{B}\|_1$ in one iterate is only LN. t ($t<5$) is the maximum search number in the inexact line search in choosing a_k to keep away from trivial search in the next iteration direction. The linear and nonlinear conjugated gradient method only has the computational complexity of $\mathcal{O}(N)$, and the complexity of vector transport in the RCG method is $\mathcal{O}(\mathrm{NL}^2)$ for updating **B**. Unlike $L_{1/2}$-NMF and GLNMF, there is no $\mathcal{O}(N^2)$ operation in our proposed algorithm, so our proposed algorithm has much lower computational complexity than the original $L_{1/2}$-NMF. Then, we can have the following proposition.

Proposition 1: With d steps of iteration, the overall computational cost of Algorithm 1 is $\mathcal{O}(\mathrm{dMNL})$.

3) Convergence Analysis: The proposed algorithm is globally convergent for **A** and **B**. Theorem 1 illustrates the global convergence of Algorithm 1 to **A**, and Theorem 2 illustrates the global convergence of Algorithm 1 to **B**. Both theorems are proved in the Appendix.

Theorem 1: The objective function (16) is globally convergent under the updating rule (18) with respect to **A**.
The proof is attached in the Appendix.

In order to guarantee the global convergence of the proposed algorithm on the oblique manifold for **B**, we carefully determine a_k and β_{k+1}. Here, we use the inexact search to find a_k and the efficient hybrid method [51] to get β_{k+1}. c_1 and c_2 can be chosen as $0<c_1<1/2$ and $c_1<c_2<1$ in this article ($c_1=0.0001$ and $c_2=0.9$). Then, the global convergence of Algorithm 1 with respect to **B** can be given by Theorem 2, and the proof is provided in the Appendix.

Theorem 2: If the series $\{\mathbf{B}_k\}$ is generated by the updating rule (22), we have

$$\lim_{k\to\infty}\inf\|\mathrm{grad}\mathcal{L}(\mathbf{B}_k)\|=0. \tag{32}$$

IV. EXPERIMENT

This section will evaluate the performance of the proposed methods. Both synthetic and real-world data are tested. Spectral angle distance (SAD) and root-mean-square error (RMSE) are adopted to measure the accuracy of the estimated endmembers and the corresponding abundances. The SAD aims to evaluate the similarity of the ground-truth lth endmember signature $\mathbf{A}_l$ and the estimation $\hat{\mathbf{A}}_l$, and it is defined as

$$\mathrm{SAD}_k=\arccos\left(\frac{\mathbf{A}_l^T\hat{\mathbf{A}}_l}{\|\mathbf{A}_l\|\|\hat{\mathbf{A}}_l\|}\right). \tag{33}$$

The RMSE aims to evaluate the performance of abundance estimate $\mathbf{S}_l$, and the ground-truth abundance matrix for lth endmember is $\hat{\mathbf{S}}_l$, which is defined as

$$\mathrm{RMSE}_l=\left(1/N\|\mathbf{S}_l-\hat{\mathbf{S}}_l\|^2\right)^{1/2}. \tag{34}$$

To verify the performance, the proposed method is compared with several widely used HU algorithms, including VCA-FCLS [9], [42], $L_{1/2}$-NMF [28], graph-based $L_{1/2}$-NMF [29], R-CoNMF [31], ALMM [16], SULoRA [17], and EndNet [21]. We conducted the EndNet method under the setting in the original paper [21], i.e., the learning rate is 10^{-3} and the batch size is 64. Since the output of SULoRA is only the abundance matrix, we use the VCA initialization result as the endmember extracted by SULoRA. The EndNet method can only initialize the endmember matrix, i.e., the weight matrix.

A. Experiments on Synthetic Data

In order to generate the synthetic data, we choose ten spectra randomly from USGS digital spectral library. According to the introduction of [55], we generate a synthetic dataset with 188 spectral bands and 3364 samples. The default endmember number is 4.

Gaussian noise with zero mean is added into the synthetic dataset having various levels of signal-to-noise ratio (SNR) as

$$\mathrm{SNR}=10\log_{10}\frac{E\left(X^TX\right)}{e^Te} \tag{35}$$

in which X denotes the original data, e represents the noise in a pixel, and E is the expectation.

1) Influence of Different Initializations: In this experiment, we present the results of SAD and RMSE by our method under different initializations, i.e., VCA-FCLS, random initialization, and Nfindr-FCLS on the synthetic dataset. The SNR in this experiment is 15 dB with four endmembers. The results can be seen from Tables II and III. It is obvious that our method gets the lowest SAD in the three common initializations. Also, the proposed method has the lowest RMSE under VCA-FCLS initialization, while EndNet gets the other two lowest mean RMSEs under other initializations. With different initializations, the two indicators stay insensitive, and random initialization performs a little worse result than those by the other two initializations. $L_{1/2}$-NMF, GLNMF, ALMM, and SULoRA rely much on a more satisfactory initialization on the synthetic dataset. Also, R-CoNMF is also insensitive to these three initializations but does not perform as good as L_1-$\mathcal{OB}$NMF.

2) Convergence Rate Comparison: We perform the experiments of NMF-based methods, including VCA-FCLS, $L_{1/2}$-NMF, GLNMF, R-CoNMF, and L_1-$\mathcal{OB}$NMF under the

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TABLE II
SAD AND STANDARD DEVIATION ($\times 10^{-2}$) WITH DIFFERENT INITIALIZATIONS IN SYNTHETIC DATA

	$L_{1/2}$-NMF		GLNMF		R-CoNMF		ALMM		SULoRA		EndNet		L_1-$\mathcal{OB}$NMF	
	mean	std	mean	std	mean	std	mean	std	mean	std	mean	std	mean	std
nfindr-fcls	7.44	±0.07	11.34	±0.01	9.51	±0.29	10.54	±1.38	12.24	-	8.70	-	**6.45**	±0.04
random	17.34	±4.13	20.32	±4.35	9.73	±0.58	27.56	±3.28	26.81	±4.35	8.81	±1.22	**7.04**	±1.27
vca-fcls	8.31	±2.43	9.76	±2.69	9.41	±1.33	9.77	±1.56	13.99	-	9.62	-	**6.41**	±1.22

TABLE III
RMSE AND STANDARD DEVIATION ($\times 10^{-2}$) WITH DIFFERENT INITIALIZATIONS IN SYNTHETIC DATA

	$L_{1/2}$-NMF		GLNMF		R-CoNMF		ALMM		SULoRA		EndNet		L_1-$\mathcal{OB}$NMF	
	mean	std	mean	std	mean	std	mean	std	mean	std	mean	std	mean	std
nfindr-fcls	9.68	±0.07	10.76	±0.01	6.22	±0.08	7.98	±0.98	9.90	-	**4.50**	-	5.59	±0.01
random	27.53	±1.10	14.72	±4.34	6.44	±0.28	36.17	±3.28	24.38	±2.64	**4.10**	±0.68	6.33	±0.01
vca-fcls	6.82	±1.06	8.34	±2.15	5.75	±0.40	9.23	±1.83	8.79	-	5.68	-	**5.53**	±0.48

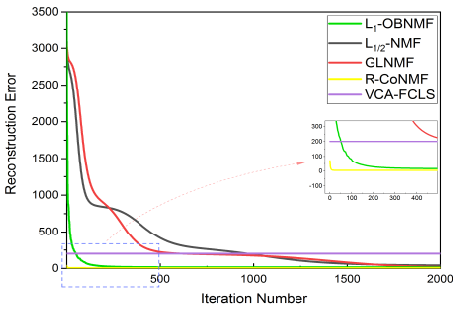


Fig. 3. Comparisons of convergence rate with state-of-the-art algorithms.

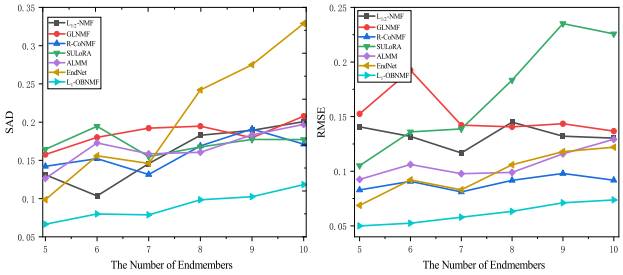


Fig. 4. Comparison of SAD and RMSE with several state-of-the-art algorithms in different numbers of endmembers.

condition of SNR = 15 dB, endmember number is 4, and random initialization. As shown in Fig. 3, the R-CoNMF has a much faster convergence rate beyond other competitors since it utilizes the observation data after dimensionality reduction. Among other algorithms, L_1-$\mathcal{OB}$NMF declines dramatically in the first 100 iterations, while GLNMF and $L_{1/2}$-NMF have to iterate more than 500 times to become stable, and this confirms our effectiveness in accelerating the convergence rate. The construction error of VCA-FCLS stays the same. At 2000 iterations, the reconstruction error of R-CoNMF reduces to 4.95, while the reconstruction errors of L_1-$\mathcal{OB}$NMF, GLNMF, and $L_{1/2}$-NMF reduce to 14.75, 20.42, and 41.47, respectively. We enlarge the local image of first 500 iterations to make the comparison more obviously.

3) Impact of Different Numbers of Endmembers: This experiment examines the impact of endmember number L to the SAD and RMSE of all the compared methods. We change the endmember number L from 5 to 10. As shown in Fig. 4, our method has the best results both for mean SAD and mean RMSE over all the compared algorithms on all the different numbers of endmembers. We can see a slight and steady increase of the black line with the rise of endmember number L both for SAD and RMSE. However, for other methods, the SAD and RMSE lines show much larger results and more "bumpy" fluctuation with the increase of L.

4) Impact of Different Noise Levels: In this experiment, we test the performance of different methods under SNR = 5, 15, 25, 35, …, ∞ dB. All the algorithms use random initialization. The left of Fig. 5 shows that L_1-$\mathcal{OB}$NMF outperforms all other NMF methods under all the noise levels in SAD, except for the mean RMSE in SNR = 35. $L_{1/2}$-NMF and GLNMF are less sensitive to noise than R-CoNMF. Turning to RMSE, as shown in the right of Fig. 5, the results of L_1-$\mathcal{OB}$NMF under different noise levels are much lower than those of other methods. RMSEs of R-CoNMF are also much lower than those of $L_{1/2}$-NMF and GLNMF. For both SAD and RMSE, the proposed method and R-CoNMF undergo a dramatic decrease when SNR increases from 5 to 15 but stay stable afterward.

B. Experiments on Real World Dataset

Here, we conduct experiments on real-world data. Two datasets are used, the Jasper Ridge Dataset and the image acquired by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor over Cuprite, NV, USA. Also, several experiments have been given in the synthetic dataset and two real-world datasets. We compare our proposed method

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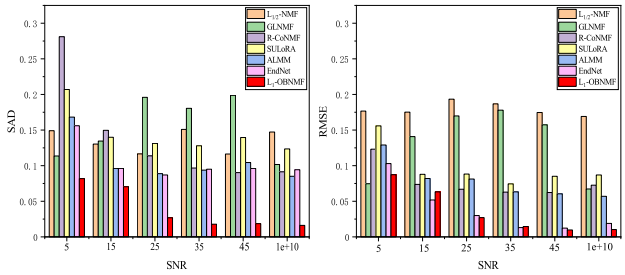
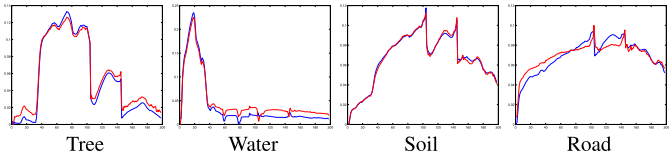


Fig. 5. Comparison of the algorithms with different SNRs in terms of SAD and RMSE.

TABLE IV

ENDMEMBER MAPS OF JASPER DATASET OF OUR METHOD AND GROUND TRUTH. THE RED LINE IS GROUND TRUTH AND THE BLUE LINE IS OUR ENDMEMBER MAP



with $L_{1/2}$-NMF, GLNMF, R-CoNMF, ALMM, SULoRA, and EndNet.

1) Jasper Ridge Dataset: The Jasper Ridge dataset is of size 512×614, with 224 spectral bands. Its spectral resolution is 10 nm acquired in the 380–2500-nm region. Since the dataset is too complicated, in most experiments, researchers use a subimage of 100×100 pixels as in [56]. After removing the low SNR bands and water-vapor absorption bands (bands 1–3, 108–112, 154–166, and 220–224), 198 bands remain to be analyzed (i.e., $M = 198$). The endmember number is 4, including soil, tree, road, and water. The ground truth of the Jasper dataset is derived from the website: http://lesun.weebly.com/hyperspectral-dataset.html. In this experiment, we adopt VCA-FCLS initialization methods mentioned above.

As shown in Table IV, in the first three endmembers, there exists high similarity on endmember map compared to the ground truth. There shows a larger difference on the reflectance of road, even our RMSE on road is relatively low. As shown in Table V, our method produces the second lowest mean SAD while EndNet gets the lowest mean SAD compared with all baseline methods. From Table VI, we can see our mean RMSE is much lower than those of other classical methods, while slightly better than the EndNet result. From Fig. 6, we get good performance on the whole, while the first abundance map looks more divergent compared with the ground truth.

2) Cuprite Dataset: The Cuprite dataset was captured by the AVIRIS sensor over Cuprite. In the Cuprite dataset, 224 bands range from 400 to 2500 nm with a size of 250×191. In most experiments, 188 bands are used after removing the low SNR bands and water-vapor absorption bands (i.e., 1 and 2, 104–113, 148–167, and 221–224 bands). According to Nascimento and Dias [9], there are 14 types of minerals in the dataset. Also, Qian *et al.* [28] set the endmember

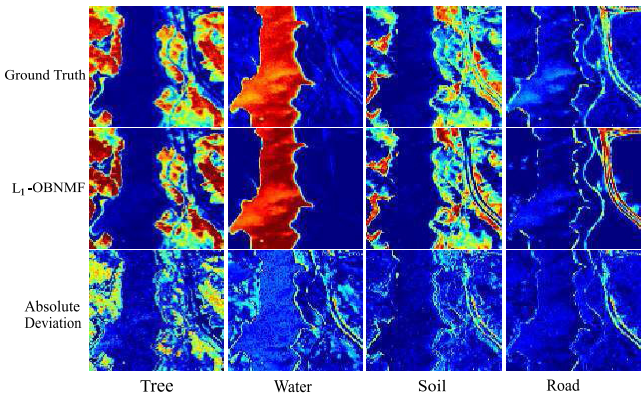


Fig. 6. Abundance maps and absolute difference map of Jasper dataset of our method and ground truth. (Top row) Our abundance map. (Middle row) ground truth. (Bottom row) Absolute difference map. (From Left column to Right) Tree, water, soil, and road.

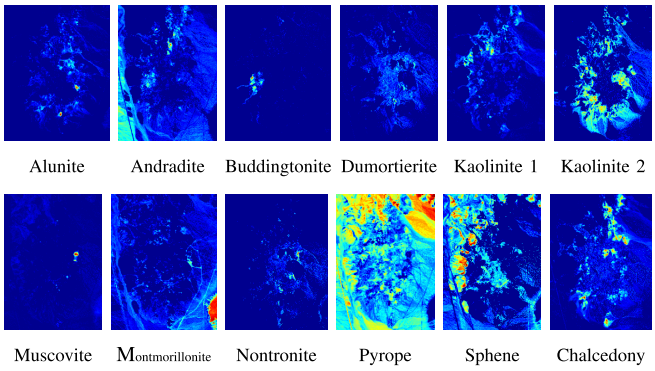


Fig. 7. Abundance maps of cuprite dataset of our proposed method.

to be 10, with high similarity on different spectra. In most cases, the number of endmembers in the Cuprite dataset is set to be 12, and we also use this setting. The ground truth of the Cuprite dataset is derived from the website: http://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes.

From Table VII, our mean SAD (in red) gets the second lowest mean SAD of all the compared methods, only 0.04 higher than GLNMF results (in bold), while much lower than R-CoNMF, ALMM, SULoRA, EndNet. $L_{1/2}$-NMF has the most lowest individual SAD of different endmembers, and our method gets two lowest individual SADs and three second lowest individual SADs. However, our superiority on computational complexity over GLNMF is obvious. There shows no apparent connection between individual SAD and mean SAD, as $L_{1/2}$-NMF gets the most number of lowest individual SADs, while its mean SAD is higher than those of GLNMF and $L_{1/2}$-NMF. Since no official abundance map is given, we only provide our own abundance map produced by our proposed algorithm. Fig. 7 shows our resulting whole abundance map. It can be seen that it reveals good sparsity in most of the abundance maps, which is congruent to our objective.

3) Comprehensive Comparison: We conducted an ablation experiment to show the effectiveness of $\mathcal{OB}$ and the L_1 regularization, as shown in Table VIII. Then, we represent the computational time of each compared method in the whole

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TABLE V

SAD AND STANDARD DEVIATION ($\times 10^{-2}$) ON THE JASPER DATASET OF DIFFERENT METHODS. THE BEST RESULTS ARE IN BLACK BOLD, WHILE THE SECOND BEST ARE IN RED

	$L_{1/2}$-NMF		GLNMF		R-CoNMF		ALMM		SULoRA		EndNet		L_1-$\mathcal{OB}$NMF	
	mean	std	mean	std	mean	std	mean	std	mean	std	mean	std	mean	std
Tree	23.81	±1.5	10.27	±4.90	17.91	±0.99	15.75	±3.36	22.86	-	**6.92**	**±0.04**	17.45	±1.38
Soil	18.28	±5.7	11.20	±0.75	34.63	±3.56	22.82	±0.05	25.23	-	**3.80**	**±0.14**	9.96	±2.39
Water	16.15	±2.9	20.43	±0.93	18.77	±0.57	69.59	±9.52	70.27	-	**5.88**	**±0.01**	7.18	±0.30
Road	35.56	±11.5	7.88	±0.78	5.15	±0.56	33.69	±17.14	44.47	-	**2.72**	**±0.03**	3.32	±0.65
Mean	23.46	±3.5	11.75	±0.89	19.11	±1.09	35.46	±2.74	40.70	-	**4.81**	**±0.12**	9.47	±0.29

TABLE VI

RMSE AND STANDARD DEVIATION ($\times 10^{-2}$) ON THE JASPER DATASET OF DIFFERENT METHODS. THE BEST RESULTS ARE IN BLACK BOLD, WHILE THE SECOND BEST ARE IN RED

	$L_{1/2}$-NMF		GLNMF		R-CoNMF		ALMM		SULoRA		EndNet		L_1-$\mathcal{OB}$NMF	
	mean	std	mean	std	mean	std	mean	std	mean	std	mean	std	mean	std
Tree	19.35	±1.9	23.97	±3.20	17.83	±0.26	29.88	±15.29	12.52	-	12.90	±0.03	**6.63**	**±0.95**
Soil	29.38	±4.2	23.72	±4.11	20.92	±0.57	37.09	±22.77	11.73	-	11.42	±0.06	**7.07**	**±1.20**
Water	22.20	±8.3	26.56	±3.77	15.47	±0.27	38.20	±0.85	16.41	-	**5.47**	**±0.11**	11.82	±0.25
Road	28.34	±4.7	12.30	±2.39	12.62	±0.54	24.01	±2.78	26.53	-	**9.22**	**±0.08**	12.34	±1.15
Mean	24.91	±3.3	21.64	±2.66	19.11	±0.24	29.65	±11.6	16.80	-	9.73	±0.17	**9.46**	**±0.86**

TABLE VII

SAD AND STANDARD DEVIATION ($\times 10^{-2}$) ON THE CUPRITE DATASET OF DIFFERENT METHODS. THE BEST RESULTS ARE IN BLACK BOLD, WHILE THE SECOND BEST ARE IN RED

	GLNMF		$L_{1/2}$-NMF		R-CoNMF		ALMM		SULoRA		EndNet		L_1-$\mathcal{OB}$NMF	
	mean	std	mean	std	mean	std	mean	std	mean	std	mean	std	mean	std
Alunite	10.03	±0.31	25.29	±0.74	12.43	±1.81	34.61	±1.13	23.85	-	18.83	±7.36	**9.65**	±0.16
Andradite	8.50	±0.12	**6.85**	**±0.73**	13.59	±1.22	7.60	±0.20	7.93	-	7.32	±1.17	10.87	±3.54
Buddingtonite	7.88	±0.94	12.98	±0.93	13.96	±2.16	10.59	±1.55	**7.50**	-	9.76	±1.91	7.58	±1.97
Dumortierite	12.87	± 0.59	**8.65**	±0.45	15.62	±4.32	9.45	±0.53	11.42	-	14.83	±4.27	11.29	±0.53
Kaolinite 1	9.08	±0.05	7.37	±0.39	18.13	±2.79	9.79	±0.36	7.47	-	13.56	±4.75	**7.09**	±1.28
Kaolinite 2	7.42	±1.15	**6.24**	±0.11	11.03	±5.44	15.65	±5.09	9.77	-	12.31	±1.24	9.71	±0.81
Muscovite	9.46	±0.25	17.24	±1.28	19.45	±3.51	17.16	±1.96	**9.42**	-	15.73	±3.23	11.27	±2.16
Montmorillonite	11.47	±1.54	**5.91**	±0.08	11.12	±2.60	11.77	±4.12	15.70	-	9.92	±2.31	7.09	±0.16
Nontronite	9.67	±0.18	7.52	±0.29	12.60	±0.61	9.82	±0.47	**7.13**	-	9.43	±0.85	8.63	±0.52
Pyrope	**5.28**	**±0.13**	5.86	±0.41	7.18	±3.93	5.59	±0.06	6.56	-	6.72	±1.47	13.60	±2.46
Sphene	9.64	±3.33	**5.88**	**±0.81**	9.36	±1.77	7.65	±0.25	7.47	-	5.99	±1.10	10.47	±2.69
Chalcedony	14.17	±0.93	12.39	±0.33	17.24	±5.82	**10.52**	**±0.45**	10.54	-	13.57	±3.27	12.89	±0.31
Mean	**9.67**	**±2.05**	10.17	±0.24	13.65	±0.61	13.57	±3.27	10.39	-	11.49	±1.47	9.71	±2.21

three datasets in Table X. We also show the effect of different initialization on the performance on the proposed method in Table IX. From Table VIII, the effectiveness of $\mathcal{OB}$ is clearly shown on the three datasets over $L_{1/2}$-NMF. However, the effect of L_1 regularization is not clear on synthetic and Jasper datasets, as the mean SADs of $\mathcal{OB}$NMF are lower than those of $L_{1/2}$-$\mathcal{OB}$NMF. Since the Riemannian optimization is directly based on the curved manifolds, we eliminate the sum-to-one constraint to obtain an unconstrained optimization problem on $\mathcal{OB}$ manifold that will only use feasible points. This allows us to incorporate its intrinsic Riemannian geometry in the resulting optimization problems, thus producing far more accurate numerical results [38], [40]. We also showed the effect of L_1 regularizer. However, in the synthetic dataset which is much simpler than Jasper and Cuprite datasets, there seldom exists any influence of L_1 regularizer. From Table IX, in the Jasper and Cuprite datasets, Vca-Fcls intialization generates much better performance than other two methods. Random initialization shows the worst results on the three datasets. Also, the results show a tendency that more complex dataset requires better initialization to get satisfying results, and moreover, the proposed method still relies much on the initialization. From Table X, we list the computational time of each compared method in different datasets to get the best results. The computational time of our method is competitive among all the compared methods. Combining

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TABLE VIII

MEAN SAD AND MEAN RMSE ($\times 10^{-2}$) OF $L_{1/2}$-NMF, $\mathcal{OB}$NMF, AND L_1-$\mathcal{OB}$NMF ON THE SYNTHETIC, JASPER, AND CUPRITE DATASETS

	Synthetic data				Jasper				Cuprite			
	SAD		RMSE		SAD		RMSE		SAD		RMSE	
$L_{1/2}$-NMF	8.31	±2.43	6.82	±1.06	23.46	±3.50	24.91	±3.30	10.17	±0.24	-	-
$\mathcal{OB}$NMF	**6.03**	±**1.68**	5.83	±1.21	**9.25**	±**0.83**	10.97	±0.23	10.08	±0.44	-	-
L_1-$\mathcal{OB}$NMF	6.41	±1.22	**5.53**	±**0.48**	9.47	±0.29	**9.46**	±**0.86**	**9.71**	±**2.21**	-	-

TABLE IX

MEAN SAD AND MEAN RMSE ($\times 10^{-2}$) OF L_1-$\mathcal{OB}$NMF UNDER RANDOM, NFINDR-FCLS, AND VCA-FCLS INITIALIZATION ON ALL THREE DATASETS

	Synthetic data		Jasper		Cuprite	
	SAD	RMSE	SAD	RMSE	SAD	RMSE
Random Init	7.04	6.33	22.47	22.23	11.82	-
Nfindr-Fcls	6.45	5.59	13.66	13.85	10.27	-
Vca-Fcls	**6.41**	**5.53**	**9.47**	**9.46**	**9.71**	-

TABLE X

COMPUTATION TIME (SECONDS) OF DIFFERENT METHODS ON THE JASPER AND CUPRITE DATASETS

	$L_{1/2}$-NMF	GLNMF	R-CoNMF	ALMM	SULoRA	EndNet	L_1-$\mathcal{OB}$NMF
Synthetic	2.88	4.99	2.15	14.40	**1.19**	-	1.83
Jasper	10.03	47.71	5.59	49.87	59.52	-	**2.83**
Cuprite	18.33	33.11	55.53	271.66	**9.05**	-	25.70

TABLE XI

MEAN SAD AND MEAN RMSE ($\times 10^{-2}$) OF DIFFERENT PARAMETERS λ OF L_1-$\mathcal{OB}$NMF ON THE JASPER AND CUPRITE DATASETS

	0	0.001	0.01	0.1	1	10
	SAD	SAD	SAD	SAD	SAD	SAD
Jasper	9.25	8.86	8.51	7.85	7.87	8.21
Cuprite	10.08	10.07	10.27	10.01	10.74	10.83
	RMSE	RMSE	RMSE	RMSE	RMSE	RMSE
Jasper	10.97	9.78	10.93	10.92	10.96	10.98
Cuprite	-	-	-	-	-	-

Tables II and III, the VCA-FCLS initialization generates the best results in the three datasets. Table XI represents the influence of the parameter λ selection in 15 in real datasets. λ is set as 0.001, 0.01, 0.1, 0, 1, 10. In the Jasper dataset, mean SAD enjoys a drop from 0 to 1, whereas mean RMSE enjoys an increase after a drop during 0 to 1. In the Cuprite dataset, the lowest mean SAD is located around 0.1.

C. Summary for Experiments

Based on the experiments on the synthetic dataset, our proposed method shows overall stability and superiority under different conditions, including endmember change and SNR change. The convergence rate is only slower than R-CoNMF but faster than $L_{1/2}$-NMF and GLNMF. For Jasper dataset, our mean RMSE surpasses other state-of-the-art methods and our mean SAD is only worse than EndNet's result. However, for Cuprite dataset, the results of GLNMF, $L_{1/2}$-NMF, R-CoNMF, SULoRA, and our methods are quite near, and our mean SAD is slightly higher than GLNMF and lower than the other five methods. From the two abundance maps in Figs. 6 and 7, our proposed algorithm gets satisfying sparsity of the abundance map on these two real datasets. The ablation experiment shows the effectiveness of the proposed method, as shown in Table VIII. The possible reasons behind might be summarized into two points. First, since the Riemannian optimization is directly based on the curved manifolds, we can eliminate the ASC and ANC constraint to obtain an unconstrained optimization problem on the $\mathcal{OB}$ manifold that the derived search space admits a natural manifold structure. This allows us to incorporate its intrinsic Riemannian geometry in the resulting optimization problems, thus producing far more accurate numerical results [38], [57]. Second, in the sense of the manifold structure, the objective function is smooth, which results in easier optimization mathematically [36]. Moreover, empirically, both for synthetic and real-world datasets, the performance of the NMF methods still relies much on the initialization.

V. CONCLUSION

In this article, a new sparse $\mathcal{OB}$NMF algorithm is proposed to improve the performance of existing HU algorithms. The proposed method not only explores the oblique-manifold structure of abundance matrix but also keeps the superiority of $L_{1/2}$ regularization with less computational cost. Moreover, this method can also be regarded as a novel way to deal with ASC and ANC. The multiplicative updating rule and RCG rule on oblique manifold in our proposed algorithm lead to a large improvement in convergence rate. Experimental results on synthetic and real-world datasets show its advantages on unmixing performance, convergence rate, and stability to noise.

VI. DISCUSSION

The proposed algorithm provides a novel perspective to deal with HU problem, by regarding the abundance matrix as lying on the oblique manifold, which will transform the common $L_{1/2}$-NMF into a problem without any constrained condition. This can also be a substitute of common FCLS method in dealing with ASC and ANC. With conjugated gradient algorithm on the oblique manifold, less complexity and faster convergence rate will be derived consequently. Since few consideration has been taken about local geometric information or pairwise connection in the proposed model, future study will focus on exploring the mutual connection of different pixels or abundance vectors.

APPENDIX

Proof of Theorem 1:

Definition 1: Regarding (16) as the function of $\mathbf{A}$, let $h(\mathbf{A}, \hat{\mathbf{A}})$ is the auxiliary function to $\mathcal{L}_{\mathbf{A}}$, if and only if $h(\mathbf{A}, \hat{\mathbf{A}}) \geq \mathcal{L}_{\mathbf{A}}$ and $h(\mathbf{A}, \mathbf{A}) = \mathcal{L}_{\mathbf{A}}$.

Since $h(\mathbf{A}, \hat{\mathbf{A}})$ is an upper bound of $\mathcal{L}_{\mathbf{A}}$, it is easy to replace the multiplicative updating rule of (18) by

$$\mathbf{A}_{k+1} = \arg\min_{\mathbf{A}} \; h(\mathbf{A}, \mathbf{A}_k). \tag{36}$$

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Let

$$\hat{\rho}_{mnL} = \frac{\hat{\mathbf{A}}_{ml}\mathbf{S}_{ln}}{\sum_{l=1}^{L}\hat{\mathbf{A}}_{ml}\mathbf{S}_{ln}} \tag{37}$$

then, $\sum_{l=1}^{L}\hat{\rho}_{mnl}=1$. Since the quadratic term of $\mathcal{L}_A$ is convex, by utilizing Jensen's inequality, we obtain

$$\begin{aligned}\mathcal{L}_{\mathbf{A}} &= \frac{1}{2}\sum_{m,n=1}^{M,N}\left(\mathbf{X}_{mn}-\sum_{l=1}^{L}\mathbf{A}_{ml}\mathbf{S}_{ln}\right)^2+\frac{1}{2}\lambda\sum_{n,l=1}^{N,L}|\mathbf{B}_{ln}|\\ &\le \frac{1}{2}\sum_{m,n,l=1}^{M,N,L}\hat{\rho}_{mnl}\left(\mathbf{X}_{mn}-\frac{\mathbf{A}_{ml}\mathbf{S}_{ln}}{\hat{\rho}_{mnl}}\right)^2+\frac{1}{2}\lambda\sum_{n,l=1}^{N,L}|\mathbf{B}_{ln}|\\ &\triangleq h\left(\mathbf{A},\hat{\mathbf{A}}\right).\end{aligned} \tag{38}$$

It is natural to see that $h(\mathbf{A},\mathbf{A})=\mathcal{L}_{\mathbf{A}}$. Therefore, $h(\mathbf{A},\hat{\mathbf{A}})$ is the auxiliary function to $\mathcal{L}_{\mathbf{A}}$.

By taking the derivative of $h(\mathbf{A},\hat{\mathbf{A}})$ to $\mathbf{A}_{ml}$, we obtain

$$\partial h\left(\mathbf{A},\hat{\mathbf{A}}\right)\Big/\partial\mathbf{A}_{ml}=\left[\mathbf{A}\mathbf{S}\mathbf{S}^T\right]_{ml}\left[\frac{\mathbf{A}}{\hat{\mathbf{A}}}\right]_{ml}-\left[\mathbf{X}\mathbf{S}^T\right]_{ml}. \tag{39}$$

For (39), when we let $\partial h(\mathbf{A},\hat{\mathbf{A}})/\partial\mathbf{A}=0$, we have the following updating rule of $\mathbf{A}_{ml}$:

$$\mathbf{A}_{ml}=\hat{\mathbf{A}}_{ml}\frac{\left[\mathbf{X}\mathbf{S}^T\right]_{ml}}{\left[\mathbf{A}\mathbf{S}\mathbf{S}^T\right]_{ml}}. \tag{40}$$

This rule is equivalent to the updating rule of (18). Thus, the global convergence of the updating rule of (18) with respect to $\mathbf{A}$ is proved. ■

Notation g_k is used to represent $grad\mathcal{L}(\mathbf{B}_k)$ for convenience afterward and k denotes the kth iteration.

Assumption 1: The objective function f is continuous and bounded below C^1-class, and there is a Lipschitzian constant $L>0$ such that

$$\begin{aligned}&\left|D\left(f\circ R_{\mathbf{S}_k}\right)(t\eta)[\eta]-D\left(f\circ R_{\mathbf{S}_k}\right)(0)[\eta]\right|\le Lt\\ &\eta\in T_{S_k}\mathcal{S},\quad \|\eta_k\|=1,\ S_k\in\mathcal{S},\ t\ge 0\end{aligned} \tag{41}$$

in which $\mathcal{S}$ denotes a unit sphere and R is retraction.

Since oblique manifold is the Cartesian product of a sphere $\mathcal{S}$, as shown in [58], the objective function $\mathcal{L}$ satisfies Assumption 1.

Proposition 2 (Zoutendijk's Theorem [47]): Assume that a set of series of $\{\mathbf{B}_k\}$ that lying on a Riemannian manifold are generated by (22). Let each searching direction Ξ_k satisfy that $\langle g_k,\Xi_k\rangle<0$, and a_k satisfies weak Wolfe condition (23) and (24). Then, if the assumption 1 is satisfied, then the following series converges:

$$\sum_{k=0}^{\infty}\cos^2\theta_k\|g_k\|^2<\infty \tag{42}$$

in which $\cos\theta_k=(-\langle g_k,\Xi_k\rangle)/(\|g_k\|\|\Xi_k\|)$.

Proposition 3: If $g_k\neq 0$, then Algorithm 1 is well defined for $\mathbf{B}$, and the following two inequalities hold:

$$\langle g_k,\Xi_k\rangle<0 \tag{43}$$

$$\langle g_k,\Xi_k\rangle<\langle g_{k+1},\mathcal{T}_{a_k\Xi_k}(\Xi_k)\rangle. \tag{44}$$

Proof: For $k=0$, $g_0=-\Xi_0$, then (43) is proved.

If $\langle g_1,\mathcal{T}_{a_0\Xi_0}(\Xi_0)\rangle\ge 0$, then $\langle g_1,\mathcal{T}_{a_0\Xi_0}(\Xi_0)\rangle\ge 0>\langle g_0,\Xi_0\rangle$.

If $\langle g_1,\mathcal{T}_{a_0\Xi_0}(\Xi_0)\rangle<0$, then $\langle g_1,\mathcal{T}_{a_0\Xi_0}(\Xi_0)\rangle<0$.

Considering (24) and (43) and $0<c_2<1$, then $\langle g_1,\mathcal{T}_{a_0\Xi_0}(\Xi_0)\rangle>c_2\langle g_0,\Xi_0\rangle\langle g_0,\Xi_0\rangle$.

Thus, when $k=0$, Proposition 3 holds.

For $k\neq 0$, suppose that the proposition holds for some $k>0$, then

$$\begin{aligned}\langle g_{k+1},\Xi_{k+1}\rangle &= \langle g_{k+1},-g_{k+1}+\beta_{k+1}\mathcal{T}_{a_k\Xi_k}(\Xi_k)\rangle\\ &= -\|g_{k+1}\|^2+\beta_{k+1}\langle g_{k+1},\mathcal{T}_{a_k\Xi_k}(\Xi_k)\rangle.\end{aligned} \tag{45}$$

Since the work [58] has proved when $\beta_{k+1}=\beta_{k+1}^{\mathrm{DY}}$, $\langle g_{k+1},\Xi_{k+1}\rangle<0$ holds. Then, (43) holds for $k+1$. Similarly, we have $\langle g_{k+1},\Xi_{k+1}\rangle<\langle g_{k+2},\mathcal{T}_{a_{k+1}\Xi_{k+1}}(\Xi_{k+1})\rangle$, and then, (44) holds for $k+1$.

The proof is completed. ■

Proof of Theorem 2:

Note that the objective function (16) with respect to $\mathbf{A}_{mk}$ satisfies the assumptions in Theorem 2, and then, we have

$$\sum_{k=0}^{\infty}\frac{<g_k\Xi_k>^2}{\|\Xi_k\|^2}<\infty. \tag{46}$$

If $g_{k_0}=0$ is satisfied for some k_0, then $\beta_{k_0}=0$. Then, $\Xi_{k_0}=0$, i.e., $B_{k_0+1}=R_{\mathbf{B}_{k_0}}(0)=\mathbf{B}_{k_0}$, so (32) holds for $g_k\neq 0$. Then, $\Xi_{k+1}+g_{k+1}=\beta_{k+1}\mathcal{T}_{a_k\Xi_k}(\Xi_k)$. Taking the norm and square of (46), we have

$$\|\Xi_{k+1}\|^2=\beta_{k+1}^2\|\mathcal{T}_{a_k\Xi_k}(\Xi_k)\|^2-2\langle g_{k+1},\Xi_{k+1}\rangle-\|g_{k+1}\|^2. \tag{47}$$

From (43), dividing (47) by $\langle g_{k+1},\Xi_{k+1}\rangle^2$, it obtains

$$\begin{aligned}\frac{\|\Xi_{k+1}\|^2}{\langle g_{k+1},\Xi_{k+1}\rangle^2} &= \frac{\beta_{k+1}^2\|\mathcal{T}_{a_k\Xi_k}(\Xi_k)\|^2}{\langle g_{k+1},\Xi_{k+1}\rangle^2}+\frac{1}{\|g_{k+1}\|^2}\\ &\quad-\left(\frac{1}{\|g_k\|}+\frac{\|g_{k+1}\|}{\langle g_{k+1},\Xi_{k+1}\rangle}\right)^2.\end{aligned} \tag{48}$$

Since $\beta_k=\max\{0,\min(\beta_{k+1}^{\mathrm{HS}},\beta_{k+1}^{\mathrm{DY}})\}$ and $\|\beta_{k+1}\|^2\le\|\beta_{k+1}^{\mathrm{DY}}\|^2$. As can be seen from [58], $\|\mathcal{T}_{a_k\Xi_k}(\Xi_k)\|\le\|\Xi_k\|$. Then

$$\begin{aligned}\frac{\|\Xi_{k+1}\|^2}{\langle g_{k+1},\Xi_{k+1}\rangle^2} &\le \frac{\|\mathcal{T}_{a_k\Xi_k}(\Xi_k)\|^2}{\langle g_k,\Xi_k\rangle^2}+\frac{1}{\|g_{k+1}\|^2}\\ &\quad-\left(\frac{1}{\|g_{k+1}\|}+\frac{\|g_{k+1}\|^2}{\langle g_{k+1},\Xi_{k+1}\rangle}\right)^2\\ &\le \frac{\|\Xi_k\|^2}{\langle g_k,\Xi_k\rangle^2}+\frac{1}{\|g_{k+1}\|^2}.\end{aligned} \tag{49}$$

As $<g_k,\Xi_k>^2\neq 0$

$$\frac{\|\Xi_k\|^2}{\langle g_k,\Xi_k\rangle^2}\le\sum_{i=1}^{k}\frac{1}{\|g_i\|^2}+\frac{\|\Xi_0\|^2}{\langle g_0,\Xi_0\rangle^2}=\sum_{i=0}^{k}\frac{1}{\|g_i\|^2}. \tag{50}$$

By contradiction, since $g_k\neq 0$, (32) is not satisfied, and then, there exists a constant $C>0$ such that $\|g_k\|\ge C$

$$\frac{\|\Xi_k\|^2}{\langle g_k,\Xi_k\rangle^2}\le\frac{k+1}{C^2}. \tag{51}$$

Then, we have

$$\sum_{k=0}^{\infty}\frac{\langle g_k,\Xi_k\rangle^2}{\|\Xi_k\|^2}\ge C^2\sum_{k=0}^{\infty}\frac{1}{k+1}\to\infty. \tag{52}$$

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This conflicts with (46), and the proof is completed. Then, Algorithm 1 is globally convergent with respect to **B**. ■

REFERENCES

[1] J. M. Bioucas-Dias *et al.*, "Hyperspectral unmixing overview: Geometrical, statistical, and sparse regression-based approaches," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 5, no. 2, pp. 354–379, Apr. 2012.

[2] Y. Wei, Y. Zhou, and H. Li, "Spectral-spatial response for hyperspectral image classification," *Remote Sens.*, vol. 9, no. 3, p. 203, Feb. 2017.

[3] Y. Zhou and Y. Wei, "Learning hierarchical spectral–spatial features for hyperspectral image classification," *IEEE Trans. Cybern.*, vol. 46, no. 7, pp. 1667–1678, Jul. 2016.

[4] Z. Ye, J. Chen, H. Li, Y. Wei, G. Xiao, and J. A. Benediktsson, "Supervised functional data discriminant analysis for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 2, pp. 841–851, Feb. 2020.

[5] B. Yang, H. Li, and Z. Guo, "Learning a deep similarity network for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 1482–1496, Nov. 2020.

[6] J. Peng, L. Li, and Y. Y. Tang, "Maximum likelihood estimation-based joint sparse representation for the classification of hyperspectral remote sensing images," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 30, no. 6, pp. 1790–1802, Jun. 2019.

[7] J. M. P. Nascimento and J. M. Bioucas-Dias, "Nonlinear mixture model for hyperspectral unmixing," *Proc. SPIE*, vol. 7477, pp. 1–8, Sep. 2009.

[8] X. Fu, W.-K. Ma, T.-H. Chan, and J. M. Bioucas-Dias, "Self-dictionary sparse regression for hyperspectral unmixing: Greedy pursuit and pure pixel search are related," *IEEE J. Sel. Topics Signal Process.*, vol. 9, no. 6, pp. 1128–1141, Sep. 2015.

[9] J. M. P. Nascimento and J. M. B. Dias, "Vertex component analysis: A fast algorithm to unmix hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 4, pp. 898–910, Apr. 2005.

[10] J. W. Boardman, F. A. Kruse, and R. O. Green, "Mapping target signatures via partial unmixing of AVIRIS data," in *Proc. Summaries 5th JPL Airborne Earth Sci. Workshop*, 1995, pp. 23–26.

[11] M. E. Winter, "N-FINDR: An algorithm for fast autonomous spectral end-member determination in hyperspectral data," *Proc. SPIE*, vol. 3753, pp. 266–275, Oct. 1999.

[12] J. M. Bioucas-Dias, "A variable splitting augmented lagrangian approach to linear spectral unmixing," in *Proc. 1st Workshop Hyperspectral Image Signal Process., Evol. Remote Sens.*, Aug. 2009, pp. 1–4.

[13] J. Li and J. M. Bioucas-Dias, "Minimum volume simplex analysis: A fast algorithm to unmix hyperspectral data," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, vol. 3, Jul. 2008, pp. 250–253.

[14] B. Somers, G. P. Asner, L. Tits, and P. Coppin, "Endmember variability in spectral mixture analysis: A review," *Remote Sens. Environ.*, vol. 115, no. 7, pp. 1603–1616, Jul. 2011.

[15] A. Zare and K. C. Ho, "Endmember variability in hyperspectral analysis: Addressing spectral variability during spectral unmixing," *IEEE Signal Process. Mag.*, vol. 31, no. 1, pp. 95–104, Jan. 2014.

[16] D. Hong, N. Yokoya, J. Chanussot, and X. X. Zhu, "An augmented linear mixing model to address spectral variability for hyperspectral unmixing," *IEEE Trans. Image Process.*, vol. 28, no. 4, pp. 1923–1938, Apr. 2019.

[17] D. Hong and X. X. Zhu, "SULoRA: Subspace unmixing with low-rank attribute embedding for hyperspectral data analysis," *IEEE J. Sel. Topics Signal Process.*, vol. 12, no. 6, pp. 1351–1363, Dec. 2018.

[18] R. Liu, L. Zhang, and B. Du, "A novel endmember extraction method for hyperspectral imagery based on quantum-behaved particle swarm optimization," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 4, pp. 1610–1631, Apr. 2017.

[19] B. Du, Q. Wei, and R. Liu, "An improved quantum-behaved particle swarm optimization for endmember extraction," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 6003–6017, Aug. 2019.

[20] R. Liu and X. Zhu, "Endmember bundle extraction based on multiobjective optimization," *IEEE Trans. Geosci. Remote Sens.*, early access, Nov. 24, 2020, doi: 10.1109/TGRS.2020.3037249.

[21] O. Savas, K. Berk, and A. G. Bozdagi, "EndNet: Sparse autoencoder network for endmember extraction and hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 482–496, Jan. 2019.

[22] Y. Deng, R. Chen, and C. Wu, "Examining the deep belief network for subpixel unmixing with medium spatial resolution multispectral imagery in urban environments," *Remote Sens.*, vol. 11, no. 13, p. 1566, Jul. 2019.

[23] W. Xia, X. Liu, B. Wang, and L. Zhang, "Independent component analysis for blind unmixing of hyperspectral imagery with additional constraints," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 6, pp. 2165–2179, Jun. 2011.

[24] A. Zare, P. Gader, and G. Casella, "Sampling piecewise convex unmixing and endmember extraction," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 3, pp. 1655–1665, Mar. 2013.

[25] Y. Zhou, A. Rangarajan, and P. D. Gader, "A Gaussian mixture model representation of endmember variability in hyperspectral unmixing," *IEEE Trans. Image Process.*, vol. 27, no. 5, pp. 2242–2256, May 2018.

[26] D. D. Lee and H. S. Seung, "Learning the parts of objects by non-negative matrix factorization," *Nature*, vol. 401, no. 6755, pp. 788–791, Oct. 1999.

[27] Z. Guo, T. Wittman, and S. Osher, "L_1 unmixing and its application to hyperspectral image enhancement," *Proc. SPIE*, vol. 7334, Apr. 2009, Art. no. 73341M.

[28] Y. Qian, S. Jia, J. Zhou, and A. Robles-Kelly, "Hyperspectral unmixing via $L_{1/2}$ sparsity-constrained nonnegative matrix factorization," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 11, pp. 4282–4297, Nov. 2011.

[29] X. Lu, H. Wu, Y. Yuan, P. Yan, and X. Li, "Manifold regularized sparse NMF for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 5, pp. 2815–2826, May 2013.

[30] Z. Shu, J. Zhou, L. Tong, X. Bai, and C. Zhao, "Multilayer manifold and sparsity constrainted nonnegative matrix factorization for hyperspectral unmixing," in *Proc. IEEE Int. Conf. Image Process.*, Sep. 2015, pp. 2174–2178.

[31] J. Li, J. M. Bioucas-Dias, A. Plaza, and L. Liu, "Robust collaborative nonnegative matrix factorization for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6076–6090, Oct. 2016.

[32] Y. Qian, F. Xiong, S. Zeng, J. Zhou, and Y. Y. Tang, "Matrix-vector nonnegative tensor factorization for blind unmixing of hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 3, pp. 1776–1792, Mar. 2017.

[33] J. Peng, W. Sun, and Q. Du, "Self-paced joint sparse representation for the classification of hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 1183–1194, Feb. 2019.

[34] J. Peng, Y. Zhou, W. Sun, Q. Du, and L. Xia, "Self-paced nonnegative matrix factorization for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 2, pp. 1501–1515, Feb. 2021.

[35] J. Yao, D. Meng, Q. Zhao, W. Cao, and Z. Xu, "Nonconvex-sparsity and nonlocal-smoothness-based blind hyperspectral unmixing," *IEEE Trans. Image Process.*, vol. 28, no. 6, pp. 2991–3006, Jun. 2019.

[36] P.-A. Absil, R. Mahony, and R. Sepulchre, *Optimization Algorithms on Matrix Manifolds*. Princeton, NJ, USA: Princeton Univ. Press, 2008.

[37] Q. Wang, J. Gao, and H. Li, "Grassmannian manifold optimization assisted sparse spectral clustering," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2017, pp. 5258–5266.

[38] X. Hong, J. Gao, and S. Chen, "Semi-blind joint channel estimation and data detection on sphere manifold for MIMO with high-order QAM signaling," *J. Franklin Inst.*, vol. 357, no. 9, pp. 5680–5697, Jun. 2020.

[39] P.-A. Absil and K. A. Gallivan, "Joint diagonalization on the oblique manifold for independent component analysis," in *Proc. IEEE Int. Conf. Acoust. Speed Signal Process.*, May 2006, p. V.

[40] S. E. Selvan *et al.*, "Descent algorithms on oblique manifold for source-adaptive ICA contrast," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 23, no. 12, pp. 1930–1947, Dec. 2012.

[41] M. Kleinsteuber and H. Shen, "Blind source separation with compressively sensed linear mixtures," *IEEE Signal Process. Lett.*, vol. 19, no. 2, pp. 107–110, Feb. 2012.

[42] D. C. Heinz and C. Chang, "Fully constrained least squares linear spectral mixture analysis method for material quantification in hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 39, no. 3, pp. 529–545, Mar. 2001.

[43] X. Dai and H.-G. Müller, "Principal component analysis for functional data on Riemannian manifolds and spheres," *Ann. Statist.*, vol. 46, no. 6B, pp. 3334–3361, Dec. 2018.

[44] V. Pawlowsky-Glahn and A. Buccianti, *Compositional Data Analysis: Theory and Applications*. Hoboken, NJ, USA: Wiley, 2011.

[45] J. C. Gilbert and J. Nocedal, "Global convergence properties of conjugate gradient methods for optimization," *SIAM J. Optim.*, vol. 2, no. 1, pp. 21–42, Feb. 1992.

[46] S. T. Smith, "Optimization techniques on Riemannian manifolds," *Mathematics*, vol. 158, no. 2, pp. 328–342, 2014.

[47] Y. Yang, "Globally convergent optimization algorithms on Riemannian manifolds: Uniform framework for unconstrained and constrained optimization," *J. Optim. Theory Appl.*, vol. 132, no. 2, pp. 245–265, Jun. 2007.

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[48] S. Hosseini and M. R. Pouryayevali, "Nonsmooth optimization techniques on Riemannian manifolds," *J. Optim. Theory Appl.*, vol. 158, no. 2, pp. 328–342, Aug. 2013.
[49] M. R. Hestenes and E. Stiefel, "Methods of conjugate gradients for solving linear systems," *J. Res. Nat. Bur. Standards*, vol. 49, no. 6, pp. 409–436, Dec. 1952.
[50] Y. H. Dai and Y. Yuan, "A nonlinear conjugate gradient method with a strong global convergence property," *SIAM J. Optim.*, vol. 10, no. 1, pp. 177–182, Jan. 1999.
[51] Y. H. Dai and Y. Yuan, "An efficient hybrid conjugate gradient method for unconstrained optimization," *Ann. Oper. Res.*, vol. 103, nos. 1–4, pp. 33–47, 2001.
[52] P. O. Hoyer, "Non-negative matrix factorization with sparseness constraints," *J. Mach. Learn. Res.*, vol. 5, pp. 1457–1469, Nov. 2004.
[53] C.-I. Chang and Q. Du, "Estimation of number of spectrally distinct signal sources in hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 3, pp. 608–619, Mar. 2004.
[54] J. M. Bioucas-Dias and J. M. P. Nascimento, "Hyperspectral subspace identification," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 8, pp. 2435–2445, Aug. 2008.
[55] L. Miao and H. Qi, "Endmember extraction from highly mixed data using minimum volume constrained nonnegative matrix factorization," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 3, pp. 765–777, Mar. 2007.
[56] F. Zhu, Y. Wang, S. Xiang, B. Fan, and C. Pan, "Structured sparse method for hyperspectral unmixing," *ISPRS J. Photogramm. Remote Sens.*, vol. 88, pp. 101–118, Feb. 2014.
[57] P.-A. Absil, R. Mahony, and R. Sepulchre, "Optimization on manifolds: Methods and applications," in *Recent Advances in Optimization and Its Applications in Engineering*. Berlin, Germany: Springer, 2010, pp. 125–144.
[58] H. Sato, "A Dai–Yuan-type Riemannian conjugate gradient method with the weak Wolfe conditions," *Comput. Optim. Appl.*, vol. 64, no. 1, pp. 101–118, May 2016.



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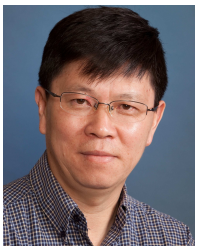
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